HDOC Day-50

AIM

To write a C program to find the next greater element for every element of a variable-length integer array using pointers, nested loops, and a user-defined function named **next_greater()**.

PROBLEM STATEMENT

Read the size and elements of an integer array **arr[]**. Pass the array to **next_greater()** using its address. For every element, search only the elements to its right and display the nearest element that is greater than the current element. Display -1 if no greater element exists to the right. Display all results in order, separated by commas.

ALGORITHM

1. Start the program.
2. Read the size of the array.
3. Read all array elements.
4. Pass the address of the array and its size to next_greater().
5. Use an outer loop to select each array element.
6. Set next = -1 for the selected element.
7. Use an inner loop to search the elements to its right.
8. When the first greater element is found, store it in next and stop the inner loop.
9. Display next for every element, separated by commas.
10. Stop the program.

C PROGRAM

```
#include <stdio.h>

void next_greater(int *arr, int size)
{
    int i, j, next;

    for(i = 0; i < size; i++)
    {
        next = -1;

        for(j = i + 1; j < size; j++)
        {
            if(*(arr + j) > *(arr + i))
            {
                next = *(arr + j);
                break;
            }
        }

        printf("%d", next);

        if(i < size - 1)
        {
            printf(", ");
```

```
        }
    }

    printf("\n");
}

int main()
{
    int size, i;

    printf("Enter size of array: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter array elements: ");
    for(i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }

    next_greater(arr, size);

    return 0;
}
```

TEST CASE TABLE

Test Case	Input Array	Expected Output
1	4 5 2 25	5, 25, 25, -1
2	13 7 6 12	-1, 12, 12, -1
3	1 3 2 4	3, 4, 4, -1
4	1 2 3 4 5	2, 3, 4, 5, -1

OUTPUT

Test Case 1

```
Enter size of array: 4
Enter array elements: 4 5 2 25
5, 25, 25, -1
```

Test Case 2

```
Enter size of array: 4
Enter array elements: 13 7 6 12
-1, 12, 12, -1
```

Test Case 3

```
Enter size of array: 4
Enter array elements: 1 3 2 4
3, 4, 4, -1
```

Test Case 4

```
Enter size of array: 5
Enter array elements: 1 2 3 4 5
2, 3, 4, 5, -1
```

RESULT

Thus, the C program to find and display the next greater element for each array element using pointers and nested loops was implemented successfully.